

Multilevel partitioning

A multilevel V-cycle coarsens the graph, partitions the tiny problem, projects the assignment back

The idea

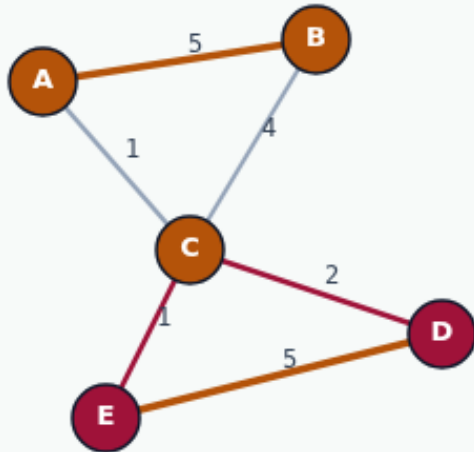
- Coarsening shrinks the search space
- The coarse partition is cheap
- Projection gives a feasible fine assignment that FM or KL can polish
- A bug in coarsening poisons every finer level
- <!-- algorithm-walkthrough -->

Coarsen with greedy merge

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Coarsen with greedy merge

Step 1 / 5 · coarsen · module04-01-multilevel-partition



Explanation

Multilevel V-cycle contracts the graph via greedy pair merge to $\text{coarseK}=2$. Supernodes $C1=\{A,B,C\}$ and $C2=\{D,E\}$ capture community structure.

- Greedy heaviest-edge merges
- Log each contraction
- Coarse graph has 2 clusters

```
coarseK=2  
labels: C1, C2
```

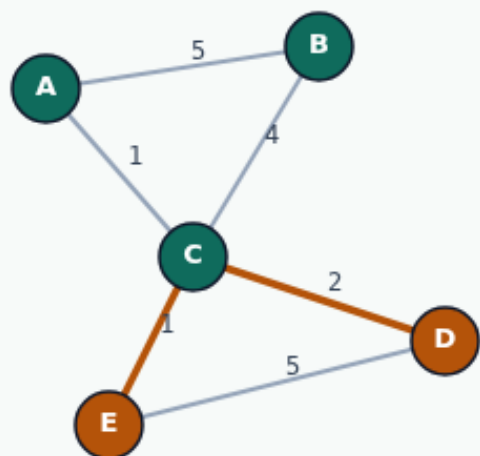
Coarsen with greedy merge

Project to fine bipartition

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Project to fine bipartition

Step 2 / 5 · project · module04-01-multilevel-partition



Explanation

Map coarse clusters to sides 0/1: $ABC \rightarrow 0$, $DE \rightarrow 1$.
Projected seed $ABC|DE$ already has cutsize 3 — unlike BAD_SEED 's cut of 12.

- Projection preserves membership
- Good global seed from coarsening
- Ready for local FM refine

```
project: ABC|DE  
cutsize: 3  
BAD_SEED cut: 12
```

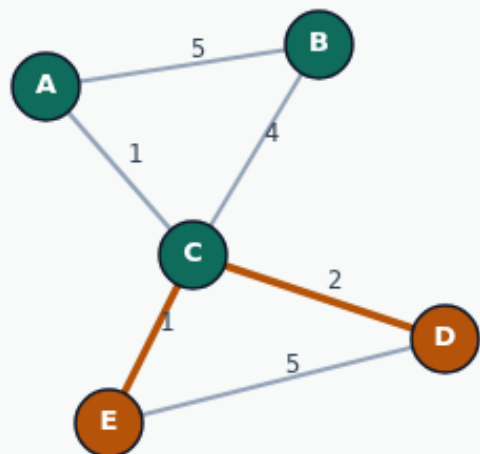
Project to fine bipartition

FM refine on fine graph

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FM refine on fine graph

Step 3 / 5 · refine · module04-01-multilevel-partition



Explanation

FM polishes the projected seed. On TINY_GRAPH the projection is already optimal — refine keeps ABC|DE at cutsize 3.

- Single-vertex moves with rollback
- Local search on fine graph
- No improvement needed here

```
refine: ABC|DE
cutsizes: 3
```

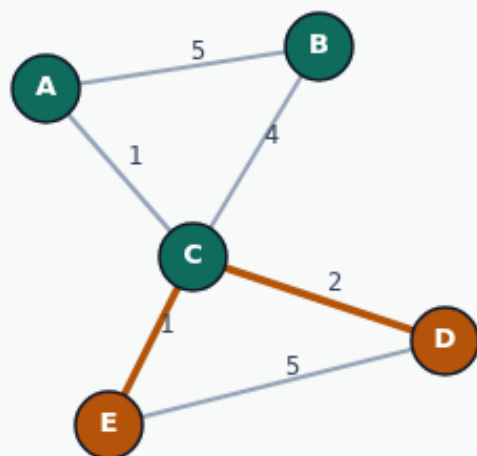
FM refine on fine graph

Final P0/P1 labels

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Final P0/P1 labels

Step 4 / 5 · final · module04-01-multilevel-partition



Explanation

Output renames sides to P0/P1 for placers. Final communities ABC|DE with cutsize 3 — multilevel beats refining a random bad seed alone.

- P0 = ABC, P1 = DE
- cutsize: 3
- V-cycle complete

```
final: ABC|DE  
cutsize: 3  
labels: P0, P1
```

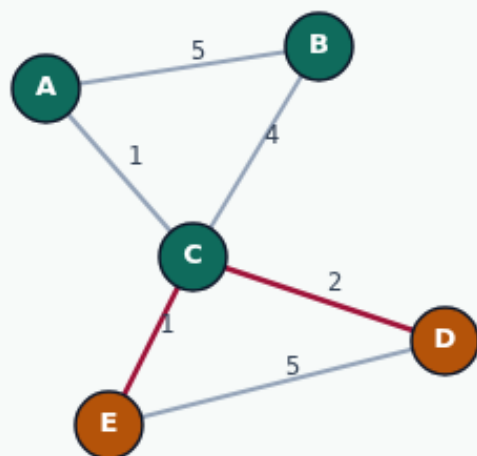
Final P0/P1 labels

Multilevel V-cycle mindset

MULTILEVEL PARTITION

Multilevel V-cycle mindset

Step 5 / 5 · takeaway · module04-01-multilevel-partition



Explanation

Coarsen for global structure, project, refine locally. Real placers nest this in repeated V-cycles with hyperedges and balance — this lab is the skeleton.

- coarsen → project → refine
- Beats BAD_SEED cut 12 → 3
- Industry default for large netlists

Starter golden: cut=3

Browser lab track

- In the browser lab track, open the **multilevel-partition** lab from the tools shelf
- Load the starter graph, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track, open this module's examples and the course `common/` solvers
- Parse the tiny graph, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps
- For multilevel flows, verify coarsening before you blame the refiner

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module